Scientific Paper Template

Design Paper for Oral Exam / Design Brief Summary

Harmonia - The hybrid tabletop game of the Museo Internazionale e Biblioteca della Musica di Bologna

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Abstract

Harmonia is a competitive hybrid phygital board game designed for the **Museo Internazionale e Biblioteca della Musica di Bologna**, developed to expand the communicative and educational potential of the museum's **Official Virtual Tour (VN 360°)**. The project responds to a set of documented pain points: the almost total absence of young visitors, the passivity of the current digital experience, the lack of agency and gamification, the limited interactivity of the collection, and the fact that the collection is significantly more enjoyable for visitors who already have a background in classical music. Through a structured human-centred design process grounded in user interviews, sentiment analysis, persona creation, UX benchmarking, co-design sessions, and the PACT framework, we designed a phygital experience that combines tangible interaction with a physical card deck and embodied interaction through haptic feedback delivered by a companion mobile app. Players take on the role of students of **Padre Martini**, competing to earn the title of Master Composer at Bologna's 18th-century Accademia Filarmonica. The project contributes three specific innovations: the structural role of haptic feedback as a primary sensory channel for musical communication, the reusability of the physical card deck across multiple narrative expansions, and a pre-visit engagement model that positions *Harmonia* as a museum gateway, bridging informal social play with formal cultural engagement.

Keywords (5–7)
Hybrid game; Museum engagement; Haptic feedback; Virtual tour; Photogrammetry; Classical music, Educational.

1. Introduction and Design Problem

1.1 Context of the project

The museum and its current state of digitalization

The project is designed for the **Museo Internazionale e Biblioteca della Musica di Bologna**, which houses the collection of musical heritage assembled by **Padre Giovanni Battista Martini**, 17,000 volumes including music books, scores, treatises, paintings and item related to the music. We examined its current state of **digitalisation**: the museum offers an **Official Virtual Tour (VN 360° project)**, developed in collaboration with the Italian Japanese communication studio *Veronesi Namioka*. This tool allows users to explore the museum's rooms remotely. The **visibility** of the Virtual Tour itself, as observed during our visit, lacks direct references inside the physical exhibition of the *Museo Internazionale e Biblioteca della Musica di Bologna*, making it difficult to discover without prior knowledge of its existence.

Strengths of the Virtual Tour

- **Accessible from any electronic device, completely free of charge**: tablets, smartphones, computers, and VR headsets.
- **Accessibility and inclusion**: it allows remote visits to the museum, removing physical, geographical, and temporal barriers.
- **Hybrid integration:** it is not intended to replace the physical visit, but to create continuity.

Interface and interaction features

- **360° Panoramic Navigation**: the user moves from room to room through an intuitive interface, clicking on visual indicators placed on the floors or doors.
- **Multi-level informative hotspots**: points of interest are positioned throughout the rooms; clicking on them reveals pop-up cards containing in depth texts and audio tracks, a fundamental addition for a music-focused museum, addressing the need to listen to what one is looking at.

Why the institution engaged our team

The Museo Internazionale e Biblioteca della Musica engaged our team as UX and Interaction Designers with the goal of **expanding** and **strengthening the communicative** and **educational** potential of the museum's Official Virtual Tour, using it as a starting point for the design of a hybrid game.

1.2 Main pain points / problems the project aims to solve

During the initial briefing with the museum and through our own research on the case study analysis, several problems emerged that the institution would like to address through the design of a hybrid (phygital) game:

1. It emerged that **most museum visitors are tourists aged over 40**, and that students and young people represent a very small proportion of visitors, despite free entry for under 18s and discounted tickets for university students.
2. **The absence of agency and gamification**: the user is a passive visitor. The Virtual Tour lacks narrative components and goals that could stimulate attention, curiosity, and learning.
3. **The collection of items does not receive adequate attention within the Virtual Tour**: they could be implemented as clickable objects, explorable in 3D through photogrammetry. The current experience is primarily architectural and spatial.
4. **Limited visibility**: during our visit to the museum, we observed that the physical exhibition lacks any direct reference to the Virtual Tour. Without prior knowledge of its existence, it is very difficult to discover it and benefit from the audio content it contains.

Pain point / problem	Who is affected?	Why does it matter for the design?
Outreach to young visitors	Museum	The game must engage and attract a younger audience that currently does not visit the museum.
Agency and gamification	Audience / Museum	The game must compensate for the lack of gamification in the Virtual Tour, supporting learning and reducing museum fatigue.
Visibility of the Virtual Tour	Museum	The game should improve the visibility of the Virtual Tour, acting as an indirect advertisement for the digital asset.
Digitized collection but not truly interactive	Audience / Museum	The game should address the fact that the collection of items cannot be interacted with or explored in the Virtual Tour.

2. Related Work, Benchmarks and Existing Solutions

2.1 Existing references, comparable projects and technologies

Since the Virtual Tour is **highly accessible** in terms of electronic devices and physical, geographical, and temporal barriers, we decided to look for references among games that can be enjoyed outside the museum context as well.

We selected *Unlock!* as our primary case study because of its hybrid gameplay (cards + app). To fully understand the dynamics of phygital flow, how physical objects and digital applications interact with each other without disrupting the user's immersive experience, we conducted a **playtesting session** of this board game. While testing it, we identified key missing elements for our concept: haptic feedback and embodied interaction. We found inspiration for these features in video games and arcade cabinets such as *Guitar Hero* and *DrumMania*, which use vibration to stimulate sensory responses. For storytelling, the representation of historical context, and the learning component, we drew inspiration from the simulation game *My Child Lebensborn - Remastered*. We also examined *Super Mario Party* for its scoring mechanics and its mix of solo and cooperative challenges.

Reference / project / technology	What it does well	Gap for your project
Unlock! — board game	Portable, phygital, app-guided progression, storyline	Haptic feedback, embodiment, persona, competition
1. Guitar Hero — video game 2. DrumMania — arcade cabinet	Vibration feedback, sensory stimulation, score system	Storyline, portability, embodiment
My Child Lebensborn Remastered — video game	Persona simulation, learning activity, historical narrative	Haptic feedback, embodiment, multiplayer, not reusable

Super Mario Party	Vibration feedback, score system, solo and cooperative challenges, avatar selection	Embodiment, storyline, no physical and digital integration

2.2 Gap statement: what existing solutions do not solve

An analysis of these references revealed a significant UX gap: none of the existing solutions successfully combines portable hybrid gameplay, embodied interaction, narrative-driven learning, and museum heritage engagement in a single experience.

A common characteristic across these references is the secondary role assigned to haptic feedback: games like *Guitar Hero* or *Super Mario Party* employ vibrations as an additional layer to increase immersion or signal a score event, if the haptic feedback is removed, the core gameplay remains entirely unchanged. Our project aims to shift the role of haptic feedback from a simple game effect to a primary and structural element of embodied interaction: making it a non-negotiable component of the experience rather than an optional enhancement.

As the table shows, educational games such as *My Child Lebensborn* and hybrid board games such as *Unlock!* each offer features that are relevant to our project, but none combines them. Our design tries to integrate the best of each reference into a single phygital system.

3. Method and Design Process

3.1 Design method used

The project followed a structured **human-centred design process**, organised in four phases:

- **Phase 1 — Briefing and context analysis:** initial briefing with the museum to understand its goals, collection, and institutional needs. This phase included an on-site visit to the museum, analysis of the *Official Virtual Tour* from multiple devices, historical research on *Padre Giovanni Battista Martini*, and a sentiment analysis of TripAdvisor reviews through XML-encoded scraping to identify visitor patterns and pain points.
- **Phase 2 — User research and requirements:** semi-structured user interviews; persona creation based on qualitative data; UX benchmarking and gap analysis of comparable projects. The phase concluded with the PACT framework, used to scope the design situation and synthesise all research findings. Co-design sessions with dot voting were used to validate and prioritise design directions collectively.
- **Phase 3 — Ideation:** based on the research findings, we developed the design concept, identified haptic feedback and embodied interaction as the key design opportunity, and explored the technologies to implement within the hybrid system.
- **Phase 4 — Envisionment and prototyping:** game loop and mechanics definition; user flowcharts and swimlane diagrams; interactive narrative prototype in Twine.

Design activity	Output produced	Where it appears in this paper
Concept design	Definition of the strategic requirements for 'Harmonia', a competitive hybrid tabletop game targeted at young people and non-expert adults, designed to transform the museum's 360° *Virtual Tour* into an active learning tool centered on Padre Martini's historical network.	Section 4
Game loop	Formalisation of the core gameplay loop, detailing the cyclical pattern of app-guided mission progression, card extraction, competitive and cooperative mini-games, and the final win-condition based on cumulative points.	Section 5
Game mechanics	Specification of core game mechanics, including card-driven musical challenges, team-based collaboration, mistake-punishment system via penalty cards, and the use of haptic feedback as the primary sensory mechanic for music simulation.	Section 5
User flowchart	Mapping of the digital onboarding process and application workflow, illustrating the step-by-step user decisions from product acquisition and module selection to the gameplay loop and exit conditions.	Section 6
Swimlane flowchart	A cross-functional Swimlane diagram detailing the chronological interaction and responsibilities between the physical components (User, Card Deck) and digital	Section 6

	layers (Companion App, 360° Virtual Tour), highlighting the mission validation and failure-state (malus) decision logic.	
User journey map	A chronological User Journey Map and narrative that tracks the player's UX, emotional shifts, and touchpoints across all game phases (from onboarding to scoring) complemented by a branched interactive storytelling structure mapped via Twine.	Section 7

It should be noted that this process does not yet include a formal evaluation of the game with the possible users, since we have not yet delivered an MVP of the game. Usability testing and iterative refinement based on user feedback represent the natural next steps of the design process.

3.2 Data/materials used: cards, museum sources, student observations, references

To maintain the highest historical accuracy and to better understand how the museum's physical and digital spaces should be integrated into our hybrid game, we conducted a series of preparatory research activities:

- **Historical research on Padre Giovanni Battista Martini:** understanding the importance of his figure in the European musical landscape of the 18th century, and his relationships with major composers, was fundamental to building the narrative foundations of the hybrid game.
- **Visit to the Museo Internazionale e Biblioteca della Musica:** this experience was essential from multiple perspectives: reinforcing historical knowledge; understanding the spatiality of the Virtual Tour and verifying its architectural accuracy firsthand; identifying key star assets present in the exhibition and understanding how to integrate them into the hybrid game; and gathering visual and aesthetic inspiration for the game's design and graphic identity.
- **Analysis of the Official Virtual Tour** of the Museo Internazionale e Biblioteca della Musica: we accessed and tested the Virtual Tour from multiple devices — smartphone, tablet, and computer, to evaluate its interface, interaction features, and limitations from different user perspectives.
- **User interviews:** semi-structured interviews conducted with young people between 18 and 25 years old, used as qualitative data to inform the persona creation process and validate the design direction.
- **Sentiment analysis of TripAdvisor reviews:** we conducted a scraping of the museum's TripAdvisor reviews and encoded them in XML format. Through a qualitative sentiment analysis of the collected data, we identified recurring patterns in visitor experiences, emotions, and frustrations. This analysis confirmed that visitors with no prior background in classical music tend to enjoy the museum significantly more when guided by contextual knowledge, and that younger audiences are almost entirely absent from the review corpus, supporting the design direction towards a pre-visit engagement tool.

4. Design Brief

Design brief element	Student answer
Title	Harmonia – The hybrid tabletop game from the "Museo Internazionale e Biblioteca della Musica di Bologna"
Type of application / game	Hybrid, competitive
Primary audience	Students, young people (15 – 25)
Secondary audience	Adults with no background in classical music (26+)
Primary persona, short description	Davide is a 17-year-old high school student attending ITIS Primo Levi in Vignola, a city near Bologna. He has never heard of the *Museo Internazionale e Biblioteca della Musica* despite living nearby. He is comfortable with smartphones and digital interactions, motivated by social dynamics and competitive gameplay, and open to discovering new content if delivered in an engaging and narrative-driven format. He would find it much more stimulating if teachers introduced new cultural concepts through a game rather than through passive classroom listening.
Secondary persona, short description	Sandra B. is a 32-year-old working professional living in Bologna. Her free time is precious; she enjoys sport, evenings with friends, and occasional cultural experiences as long as they are accessible and not intellectually overwhelming. She is not passionate about classical music but recognizes some famous melodies. She plays board games regularly and is open to cultural experiences that spark her curiosity naturally without requiring prior expertise.
Project focus / context	Expanding and strengthening the communicative and educational

	potential of the museum’s existing Official Virtual Tour, using it as a starting point for the design of a hybrid game.

Star asset	Short description	Media / image placeholder
Padre Martini and his collection	*Padre Giovanni Battista Martini* is not simply the founder of the museum's collection; he is one of the most influential figures in 18th-century European music history. Known across the continent as the "Maestro dei Maestri", he assembled over 17,000 volumes of music books, scores, and treatises, building what became one of the most important music libraries in Europe. His figure is the narrative and historical heart of the museum, and the starting point of the entire game's experience.	
The network of great composers	Padre Martini was personally connected to the greatest composers of his era: Mozart came to Bologna at the age of 14 to pass the rigorous admission exam of the *Accademia Filarmonica* under Martini's guidance, Bach corresponded with him and held him in the highest regard, Rossini spent years in Bologna and was deeply influenced by the city's musical culture. This network of relationships transforms the museum from a local institution into a crossroads of European musical history.	 

Institution / client	Institutional goals	Audience goals
Museo Internazionale e Biblioteca della Musica di Bologna	**Increase outreach,** reaching new and underrepresented audiences, particularly young people and students, who currently do not visit the museum despite free and discounted entry policies. **Extend knowledge about heritage** using the museum's	**Boost Enchantment,** players experience a sense of wonder and emotional engagement with the story of the collection and its historical characters. **Extend Knowledge about Heritage, players** build

	Virtual Tour as an active learning tool, turning the game into a knowledge-building experience that introduces players to the history of classical music and the figures of the collection. **Improve visibility of the Official Virtual Tour;** the game acts as an indirect discovery tool for the digital asset, which currently lacks direct references inside the physical exhibition.	basic knowledge of 18th-century music history and the museum's collection through narrative-driven gameplay. **Stimulate Sensory Responses;** the game activates sensory engagement through haptic feedback and tangible interaction with the physical card deck.

Audience	Motivations	Barriers	Capabilities	Devices
Student/Young people	Wonder Time Travel Entertainment	Overlooked Group Irrelevant	**Digitisation**: highly comfortable with apps and digital interactions. **Navigation**: confidence in navigating digital environments and virtual spaces. **Mixed Reality**: familiar with hybrid physical-digital experiences.	Tablet Smartphone VReality Physical cards
Adults 26+	Wonder Time Travel Curiosity	Irrelevant Digital complexity	**Digitisation**: may need more time to familiarise with the app interface and game flow. **Navigation**: the Virtual Tour may represent their first experience with an immersive digital environment. **Mixed Reality**: open to hybrid experiences but may require clearer onboarding and guidance.	Tablet Smartphone VReality Physical cards

5. Design: Concept and Mechanics

5.1 Main idea of the project / user experience description

Harmonia is a competitive tabletop game for 1 to 6 players, designed to be **played outside** the museum to enrich the *Official Virtual Tour of the Museo Internazionale e Biblioteca della Musica di Bologna*. The game enhances the museum's collection of items by making them tangible, explorable, and narratively meaningful, compensating for the limitations of the current Virtual Tour, where objects cannot be interacted with or examined. It is conceived both as a pre-visit preparation tool, building basic knowledge and sparking curiosity before entering the museum, and as a post-visit reinforcement experience, giving continuity to what players have seen in person and deepening their connection with the collection. Players take on the role of students of *Padre Martini*, the legendary 18th-century music teacher and counterpoint expert, competing to earn the coveted title of Master Composer at Bologna's *Accademia Filarmonica*.

In particular, the narrative arc moves through several key moments:

- An introduction by Padre Martini,
- a series of musical missions to prove the players' worth,
- an encounter with the spirit of Johann Sebastian Bach in the museum's library,
- a meeting with the young Mozart,
- and a final counterpoint exam that determines the winner.

The game is designed to be easily portable, it is composed by a deck of cards and a companion mobile app.

The physical card deck

It is stored in a portable fabric pouch and includes three types of cards: *museum cards*, *malus cards*, and *theory cards*. Museum cards feature reproductions of actual objects from the collection, each with a QR code linking to the Official Virtual Tour, where 3D photogrammetry models of the objects are planned to be implemented, allowing players to explore the collection items up close and at 360 degrees for the first time.

The companion mobile app

The game unfolds through a series of missions narrated by the companion mobile app, which guides players step by step through the story. Missions alternate between individual competitive challenges and cooperative team tasks, creating a dynamic and varied experience.

We have revisited the concept of **haptic feedback**: it plays a primary structural role in the experience. The smartphone vibrates to communicate musical information, note durations, rhythms, and musical patterns, that players must interpret and reproduce during key missions.

According to our personas, the technical aspects of our game could become a barrier and derail the gaming experience before it even begins. For this reason, in addition to the minimal physical components (a deck of cards) and the app's step-by-step guide, we decided to focus on a game loop and gameplay mechanics that are simple and easy to understand right from the first playthrough.

Game loop

Players will complete various missions one by one guided by the application that tells them which card extract and how to use them. Every mission can give a certain amount of point, in competitive minigame the winner will receive the maximum number of points. In squad based minigame the winner team receive the maximum number of points. The application tells if the mission is completed or not and if the player must draw a penalty card. At the end of the game the player with the maximum number of points wins.

Game mechanics

Every minigame is based on music and use cards as a physical support to complete the challenge. Some minigames will be in team so this helps to strengthen the communication between players and collaboration, but at the end only one will be the winner. Another mechanic is the use of penalty cards to give points penalties to players and punish errors.

5.2 Winning/completion condition and scoring system

The game ends after the final "counterpoint mission". Throughout the game, players accumulate white beans (points) for successful missions and black beans (penalties) for errors and malus cards. The scores are secretly tracked by the app and revealed only at the end, when the digital ballot box displays the results of one bean at a time, recreating the suspense of the historical *Accademia Filarmonica* voting ritual. The player with the most "white beans" is crowned *Master Composer of Bologna's Accademia Filarmonica*.

6. User Flow, Interaction and Hybrid System

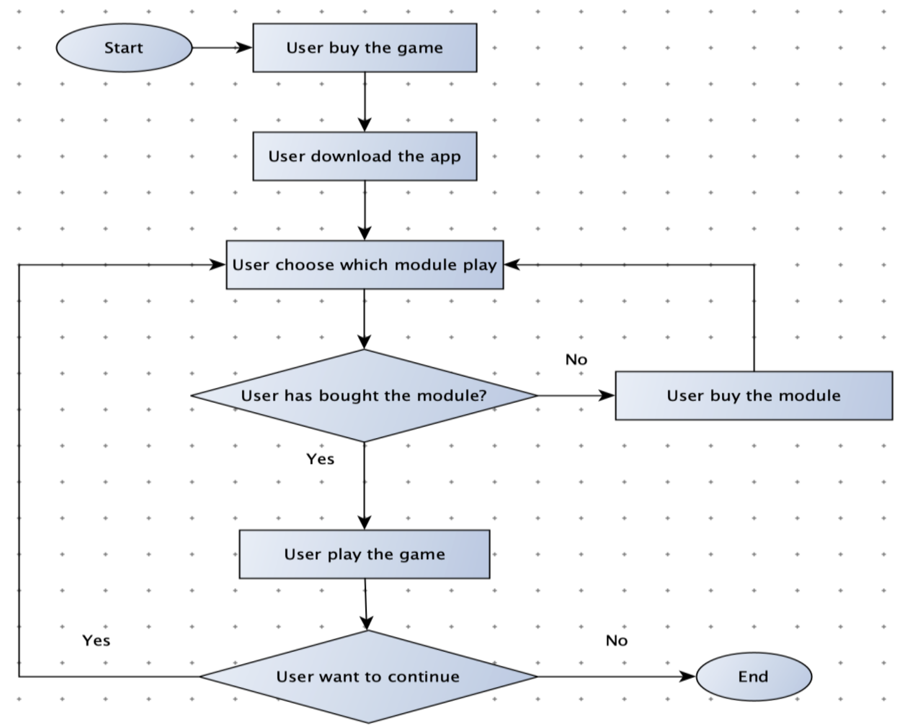
6.1 General user flow description

The flow starts with Start. First, the user buys the game, then downloads the app. Once inside the app, the user chooses which module (or expansion) they want to play. At this point, there is a decision: has the user already bought the selected module? If the answer is No, the user is directed to buy the module.

After purchasing it, the flow returns to the module selection step, allowing the user to choose a module again. If the answer is Yes, the user can access the content and play the game.

After playing, there is another decision: Does the user want to continue? If the answer is Yes, the flow returns to the module selection step, so the user can play another module or replay a module. If the answer is No, the flow ends with End.

Figure 1. General user flow diagram / flowchart



6.2 Swimlane flowchart description: actors, responsibilities, decisions and failure states

This swimlane flow is divided into three main **actors**: *the User*, the *Companion App*, the *Card Deck* and *the Virtual Tour*.

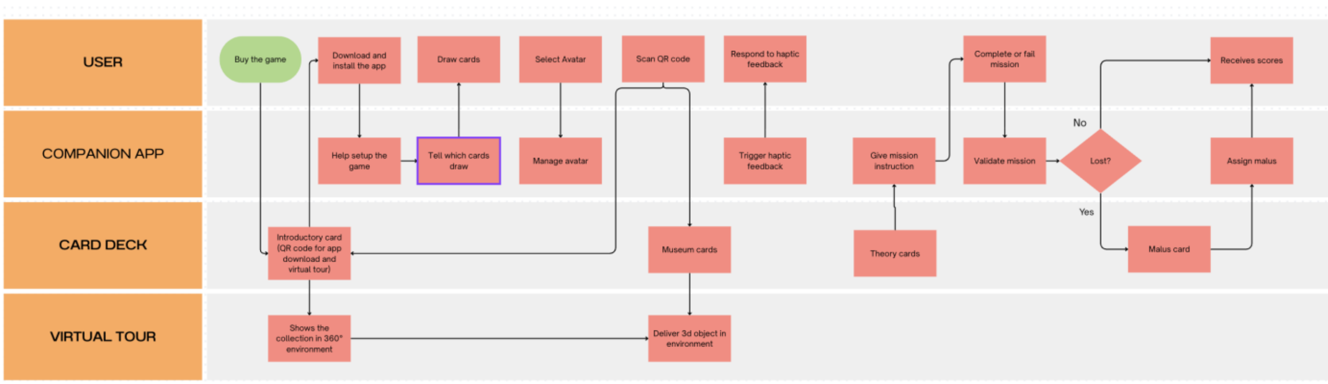
The User is responsible for the physical and interactive actions of the game. They buy the game, download and install the app, draw the required cards, select an avatar, respond to haptic feedback, scan QR codes, complete or fail missions, and finally receive their score.

The Companion App supports and manages the digital part of the experience. It helps the user set up the game, tells them which cards to draw, manages the selected avatar, triggers haptic feedback, gives mission instructions, validates the mission result, checks whether the player has lost, assigns any malus, and calculates or displays the final score.

The Card Deck provides the physical game materials used during the experience. It includes the introductory card with the QR code to download the app, museum cards, theory cards, and malus cards. These cards are connected to the app through QR codes and game instructions. And when the user scans, the qr code can see the object in 3d or the entire collection in a 3d environment.

The main decision point happens after the app validates the mission: has the user lost?
If the answer is No, the user continues and receives their score.
If the answer is Yes, the user receives a malus card, the app assigns the penalty, and then the user receives their score.

Figure 2. Swimlane flowchart



6.3 Interaction description and interaction categories

The game interaction is based on a hybrid physical-digital experience. The user interacts with both the card deck and the companion app. The cards provide the physical game content, while the app guides the player through setup, module selection, avatar management, missions, validation, penalties, and scores. While the player plays the game, he can do various missions and receive feedback from the app to see if a mission is successful or not. When the player scans the QR code, he can see the object in 3D and visualize the entire collection in the Virtual Tour.

6.4 Hybrid and technological layers

The game is built on a hybrid system that combines physical and digital elements. The physical layer is represented by the card deck, including introductory cards, museum cards, theory cards, and malus cards. These cards support the tangible gameplay experience.

The digital layer is represented by the companion app, which manages setup, module selection, avatar selection, mission instructions, validation, penalties, and scoring. The technological connection between the two layers is created through QR codes, which allow the app to link physical cards to digital content contained in the virtual tour.

Layer	Software / hardware / material	Function
Physical layer	Card	Give penality, use for minigames, give theory insight
Digital layer	Smartphone / Companion App	Deliver the narrative, controls the game flow, compute points, communicate music via vibration and haptic feedback
Museum layer	QR Code / Virtual Tour	Visualize 3d object, visualize the collection in 360° panorama
Social layer	Interaction between people	Spread the game knowledge

7. User Experience as User Journey

7.1 User journey narrative

Davide's teacher assigns *Harmonia* as a homework activity, an alternative way to explore the history of music and the cultural heritage of Bologna before a planned school trip to the *Museo Internazionale e Biblioteca della Musica*. His first reaction is mild skepticism: classical music is not something he thinks about, but the competitive premise and the simple setup reassure him. He calls some friends and they decide to play together that same evening. He does not feel intimidated: one deck of cards and an app on his phone. That he can handle.

While onboarding, *Padre Martini* appears on screen and starts telling the story of the *Accademia Filarmonica*. Davide is not fully listening; he is more interested in choosing his avatar and beating his friends. But something in the tone of the narrative catches him. It feels like a story, not a lesson.

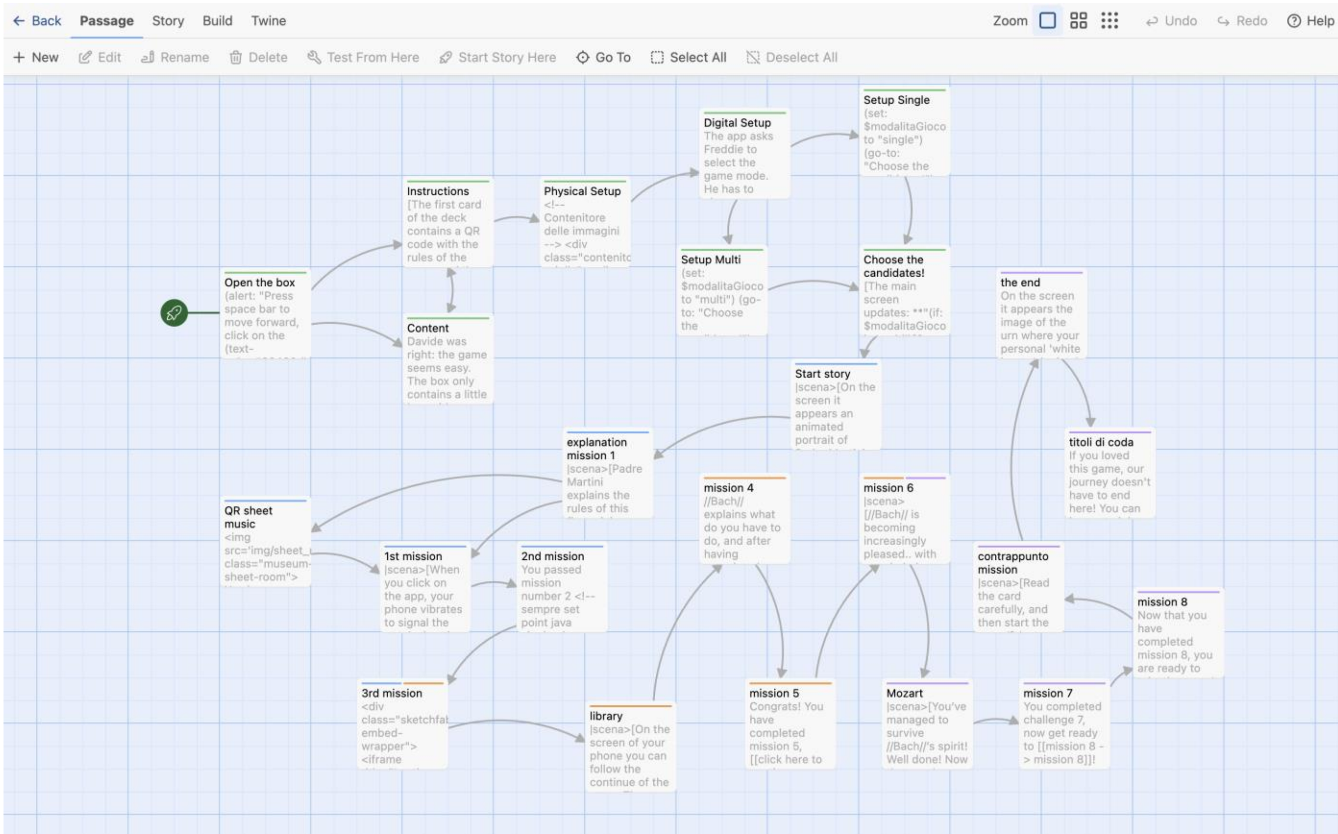
As the missions unfold, Davide starts engaging with the physical cards. When he draws a museum card and scans the QR code, a 3D object appears on his screen, a real manuscript from the collection. He pauses. "Wait, this is actually in Bologna?" The game has done something unexpected: it has made him curious without him noticing.

The haptic missions are the moment of highest engagement. His phone vibrates in patterns he has to interpret and reproduce. He laughs, fails, and tries again. His body is part of the game now, not just his eyes and his brain.

The scoring system keeps everyone tense until the end. Nobody knows who is winning. When the ballot box reveals the results of one bean at a time, the room goes quiet. The historical ritual lands even without any prior knowledge of the Accademia Filarmonica, it just feels dramatic and right.

The next day at school, Davide and his classmates talk about the game before the teacher even starts the lesson. When the school trip to the museum finally happens, Davide already knows who *Padre Martini* is, that *Mozart* studied in Bologna, and which objects he is about to see in person. The visit feels like a reunion with something he has already started to understand.

Figure 3. User Journey Map / UX diagram



Phase	User action	Touchpoint	Emotion / expectation	Pain point	Design response
Before visit / discovery	Buy the game	Users buy the game to know more about music	Excited by the new	Museum didn't give enough information about music	The box attire people because tells that improve music knowledge
Onboarding	Open the box	The player opens the box and observes the content	Curiosity	Where to place components	The game is portable and can be played anywhere
Mission / Goals	Follow apps instructions	Player complete mission or fail them	Engaged in activity	Mission not so clear	The app explains the mission until is clear
Capture / interaction	Interact with cards to complete missions	The player uses cards and smartphone to complete missions	Sense of Wonder	Cards could be used one time	Cards are general and could be reused
Feedback / scoring	Interact with the app to validate the missions	The app tracks the final scores of the players and show if they do correctly or not	Exciting because complete a mission	Feedback not clear	The app explains in deep why a mission in being complete or failed
After experience	Interact with QR code	The QR help to learn new things	Pleased because learn new thing	Virtual tour is not so informative	The theory cards have concept that teach the music

Interactive story / Twine link

https://alessandro-rocchi.github.io/projectMediaDesign/

8. Response to Pain Points and Innovation

Pain point identified in Section 1	Design solution	Expected effect
Outreach to young visitors	*Harmonia* reaches young people in their natural social contexts, at home, with friends, or through school assignments, before they ever enter the museum. The competitive and narrative-driven format transforms the museum's content into something accessible and fun, removing the perception that classical music is "not for them".	Increased awareness of and interest in the museum among 15–25-year-olds; potential adoption in school settings as a structured learning tool connecting classroom education with local cultural heritage.
The collection is more enjoyable for visitors who already have a background in classical music	The game acts as a gentle introduction to the museum's key characters, objects, and historical context, building the basic knowledge needed to appreciate the collection before the visit. As confirmed by TripAdvisor reviews, visitors who enter the museum with some contextual knowledge whether through a guide or an audio guide tend to enjoy the experience significantly more. *Harmonia* replicates this mediation effect in a playful and accessible format, making the collection enjoyable for everyone, regardless of their musical background.	Visitors of all backgrounds enter the museum with a shared baseline of knowledge, reducing the gap between expert and non-expert audiences and making the collection more accessible and meaningful for everyone.
Agency and gamification	The game replaces the passive exploration of the *Virtual Tour* with active, goal-oriented missions that alternate between individual competitive challenges and cooperative team tasks. The secret scoring system, malus cards, and haptic-driven missions maintain engagement throughout the experience.	Players develop an active relationship with the collection content, reducing museum fatigue and supporting learning through play rather than passive consumption.
Visibility of the Official Virtual Tour	Museum cards include QR codes linking directly to the Official Virtual Tour. The game acts as an indirect discovery tool for the digital asset, which currently lacks any direct reference inside the physical exhibition.	Players encounter the Official Virtual Tour naturally during gameplay, increasing its visibility and usage among audiences who would never have discovered it independently.
Digitised collection but not truly interactive (not digital)	Museum cards feature reproductions of actual collection objects with QR codes linking to planned 3D photogrammetry models within the Official Virtual Tour, allowing players to explore items up close and at 360 degrees for the first time.	The collection becomes tangible, explorable, and narratively meaningful, compensating for the purely architectural nature of the current Virtual Tour experience.

8.1 Innovation beyond the state of the art

Harmonia introduces **several specific innovation**s that go beyond the existing state of the art in hybrid museum games and digital heritage experiences.

The first innovation of **Harmonia** lies in transforming cultural learning into an active, social, and memorable **learning-by-doing** experience. Far from being a mere information tool, the game acts as a gateway to the museum: players don't passively absorb knowledge about Bologna's heritage but naturally build it through narrative immersion and friendly competition.

Another innovation is the **reusability of physical components**. The same card deck can be used for multiple story expansions, downloadable through the app. This makes the game economically sustainable, environmentally conscious, and capable of generating repeated engagement with the museum's collection over time.

The **pre-visit engagement model**. *Harmonia* is a hybrid heritage game explicitly designed to be played outside the museum as a narrative and improve the virtual tour contents. By introducing key characters, objects, and historical context before the visitor enters the museum, the game reduces cognitive overload during the visit itself, since part of the information has already been processed in a playful and emotionally engaging context, and transforms a potentially passive experience into a reunion with content already meaningfully encountered.

Another important aspect is the role of haptic feedback. Note durations, rhythms, and musical patterns are translated into tactile signals that players must physically interpret and reproduce. This repositions the smartphone from a narrative delivery tool to an embodied musical instrument, creating a genuinely new form of phygital interaction in the heritage context.

9. Conclusion

Harmonia responds to a concrete and documented set of problems identified by our team at the *Museo Internazionale e Biblioteca della Musica di Bologna*: the almost total absence of young visitors despite free and discounted entry policies, the passivity of the existing *Official Virtual Tour*, the lack of agency and gamification, the limited interactivity of the collection in digital form, and the poor visibility of the Virtual Tour inside the physical exhibition. A further critical issue, confirmed by our sentiment analysis of TripAdvisor reviews, is that the collection is significantly more enjoyable for visitors who already possess a background in classical music, a barrier that *Harmonia* directly addresses by acting as a gentle and playful introduction to the museum's key characters, objects, and historical context.

Through a structured human-centered design process, grounded in user interviews, sentiment analysis, persona creation, UX benchmarking, co-design sessions, and the PACT framework, we designed a phygital hybrid game that bridges the museum's digital and physical heritage offer. The project's main contributions are: implemented a learning by doing model that transform the activity of learn new concept in a moment of collaboration and entertainment, the pre-visit engagement model that positions *Harmonia* as a museum gateway rather than a museum tool in addition Harmonia represent a tool that can be used to strengthen the concept learned during the museum visit. Speaking of long-term engagement, Harmonia features reusable cards and extended replayability through future expansions, each introducing new themes connected to the museum and the world of music.

From a personal standpoint, we are proud of the collaborative process that led to Harmonia and of what we have achieved as a team. The project has enabled us to gain significant knowledge and develop a versatile skillset for the future. Naturally, our goal would have been to deliver a complete MVP and to conduct a formal evaluation phase with real users, and iterating based on their feedback. Usability testing, iterative refinement based on user feedback, and the full implementation of 3D photogrammetry models within the Official Virtual Tour represent the natural next steps of the design process.

Acknowledgments and Team Roles

Team member	Role in project design	Role in writing the paper
Maria Cavaleri	Research, Interactive narrative design (Twine); visual and graphic design of the game	Sections 1,2, 5, 7
Asia Marselli	User research; semi-structured interviews; persona creation; PACT framework; design brief	Sections 1, 2, 3, 4, 5, 8, 9
Alessandro Rocchi	Technical consultant; interactive narrative development (Twine); 3D modelling (Blender); user flow and swimlane diagrams; web page setup; game loop design	Sections 1,2, 5, 6, 7
Adriana Monte	User research; interviews; persona creation; sentiment analysis of TripAdvisor reviews; PACT framework; design brief	Sections 1, 3, 4, 5, 8, 9

AI Use

The authors used AI-assisted tools (Claude by Anthropic) to support the writing and creation of imagines. All content, design decisions, and research findings are the result of the authors' own work. The AI was used as a writing/graphic support tool and not as a source of data or design decisions.

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Appendix: Design Brief Summary

Our project, **Harmonia**, is a competitive phygital tabletop game combining a physical card deck with a companion mobile app, designed for the *Virtual Tour of the Museo Internazionale e Biblioteca della Musica di Bologna*.

Its primary audience is young people and students aged 15–25, and its secondary audience is adults with no background in classical music aged 26 and above.

Their characteristics are as follows. Our primary persona is Davide M., a 17-year-old high school student from Vignola, near Bologna, who has never heard of the museum despite living nearby, is comfortable with smartphones and digital interactions, and needs a narrative-driven and competitive format to feel motivated to engage with the collection. Our secondary persona is Sandra B., a 32-year-old working professional living in Bologna, who enjoys board games and occasional cultural experiences as long as they are accessible and do not require prior expertise, and is open to discovering new content when it sparks her curiosity naturally.

The context of our project is the expansion and strengthening of the communicative and educational potential of the museum's Official Virtual Tour (VN 360°), using it as a starting point for a hybrid game that acts as both a pre-visit preparation tool and a post-visit reinforcement experience.

The main pain points our project aims to solve are: the almost total absence of young visitors despite free and discounted entry policies, the passivity of *the Official Virtual Tour* and the lack of agency and gamification, the fact that the collection is significantly more enjoyable for visitors who already have a background in classical music, the limited interactivity of the collection within the digital experience, and the poor visibility of the Official Virtual Tour inside the physical exhibition.

Analyzing what has been already published and produced, and considering what is already available as a solution, we have found good references in *Unlock!*, a portable hybrid board game with app-guided progression and storyline*, Guitar Hero* and *DrumMania*, which use vibration feedback and sensory stimulation, *My Child Lebensborn Remastered*, which offers persona simulation, historical narrative, and learning activity, and *Super Mario Party*, which combines vibration feedback, a score system, and solo and cooperative challenges with avatar selection.

Nevertheless, the solutions already available do not solve the following gaps: none of the references successfully combines portable hybrid gameplay, embodied interaction, narrative-driven learning, and museum heritage engagement in a single experience, and all of them treat haptic feedback as a secondary layer (removing vibration from Guitar Hero or Super Mario Party leaves the core gameplay entirely unchanged) rather than as a primary structural element of the experience.

The star assets of our project are Padre Martini and his collection, and the network of great composers. His figure is the narrative and historical heart of the museum and the starting point of our entire game experience. The network of great composers: *Mozart*, who came to Bologna at 14 to pass the admission exam of the *Accademia Filarmonica* under Martini's guidance, Bach, who corresponded with him, and Rossini, who spent years in Bologna deeply influenced by its musical culture, transforms the museum from a local institution into a crossroads of European musical history.

The reference institution is the Museo Internazionale e Biblioteca della Musica di Bologna, and its institutional goals are to increase outreach by reaching new and underrepresented audiences, particularly young people and students, through an engaging hybrid experience, and to improve the visibility of the Official Virtual Tour, which currently lacks any direct reference inside the physical exhibition, by using our game as an indirect discovery tool.

The audience goals our project is targeting are Boost Enchantment, through which players experience a sense of wonder and emotional engagement with the collection and its historical characters, and Extend Knowledge about Heritage, through which players build basic knowledge of 18th-century music history and the museum's collection through narrative-driven gameplay.

Our primary target audience would be motivated to experience our project by wonder, the sense of surprise and discovery triggered by the narrative and the historical characters; and entertainment, the desire to have fun with friends in a competitive and socially engaging format. It might experience specific barriers like young people do not feel addressed by the museum's communication and do not perceive the space as designed for them. The audience is nevertheless capable of navigating digital environments and virtual spaces with confidence, as most of them have already experienced immersive or 360° digital environments through gaming, social media, or educational tools. Interacting with hybrid physical-digital systems such as QR codes and companion apps, represent familiar interactions in their daily lives, and engage with competitive and social gameplay formats across both physical and digital platforms. It can use devices such as smartphones, tablets, and physical cards, and these have been used as baselines for our design.

Our secondary target audience would be also motivated to experience our project by wonder, the unexpected emotional engagement triggered by discovering a historical story they did not know, and curiosity, the desire to explore something new. It might experience specific barriers like Digital Complexity, as adults may feel overwhelmed by the number of digital steps required to manage the hybrid system simultaneously, and Irrelevant, as the museum's content appears to require prior expertise that they do not possess, making them self-exclude before even engaging. The audience is nevertheless capable of navigating digital environments and virtual spaces, although for some of them the Official Virtual Tour may represent their first experience with an immersive 360° digital environment, interacting with hybrid physical-digital systems such as QR codes and companion apps, although scanning a QR code and familiarizing with the app interface may require slightly more time, and engaging with narrative and cultural content when the experience is welcoming and does not require prior expertise. It can use devices such as smartphones, tablets, and physical cards, and these have been used as baselines for our design.

The main idea of our project is a competitive narrative hybrid game in which players become students of Padre Martini, competing through a series of musical missions that introduce the key characters and objects of the museum's collection, building basic knowledge and sparking curiosity before or after a physical visit to the museum. The general user flow is described in Section 6.1 and illustrated in Figure 1.

To better describe our project, this is the User Journey: Davide receives *Harmonia* as a school assignment and plays it at home with friends. During onboarding, Padre Martini introduces the story of Accademia Filarmonica. As missions unfold, Davide draws museum cards, scans QR codes to explore real collection objects in 3D and interprets haptic signals from his smartphone during musical challenges. The secret scoring system maintains tension until the final ballot box reveals, recreating the suspense of the historical Accademia Filarmonica voting ritual. The following day at school, Davide already knows who Padre Martini and Mozart are. When the school trip to the museum happens, the visit feels like a reunion with something he already started to understand. The full user journey is illustrated in Figure 3 and accessible as an interactive narrative prototype in Twine at https://alessandro-rocchi.github.io/projectMediaDesign/

The interaction includes tangible interaction with the physical card deck; embodied interaction through haptic feedback as a primary sensory channel for musical communication, mobile interaction through the companion app managing narrative flow, avatar selection, mission instructions, validation, and scoring, and social interaction through multiplayer synchronization via Wi-Fi or Bluetooth enabling competitive and cooperative dynamics among players. The interaction diagram is illustrated in Figure 2.

The hybrid and technological elements and their role are organized across four layers. The physical layer, represented by the card deck including museum cards, malus cards, and theory cards, which provides tangible gameplay components and QR code access points. The digital layer, represented by the companion app, which delivers the narrative, controls the game flow, and computes points. The museum layer, represented by the Official Virtual Tour and planned 3D photogrammetry models, which allows players to visualize collection objects in 360° and explore them up close. The social layer, represented by multiplayer synchronization via Wi-Fi or Bluetooth, which enables competitive and cooperative dynamics among players.

Our project addresses the above mentioned pain points by reaching young audiences through a competitive and narrative-driven game playable in informal social contexts and school settings before they ever enter the museum, replacing the passivity of the Virtual Tour with active, goal-oriented missions, building basic knowledge of the collection before the visit through playful narrative mediation, making it enjoyable for everyone regardless of musical background, linking museum cards directly to the Official Virtual Tour through QR codes to increase its visibility, and implementing planned 3D photogrammetry models to make the collection truly interactive and explorable for the first time.

Its innovation lies in the pre-visit engagement model that positions Harmonia as a museum gateway: rather than simply providing information about the collection, our game transforms cultural learning into an active, social, and memorable experience through a learning-by-doing approach, allowing players to build knowledge of the museum's heritage naturally through play, competition, and narrative immersion, leaving a lasting and positive memory of the experience. Once the game has achieved its educational goal, it does not end there: players can purchase new narrative expansions, each exploring different items and storytelling from the collection not covered in the base game. This reusability and expandability of the physical card deck make Harmonia economically sustainable and capable of generating repeated and deepening engagement with the museum over time, transforming a one-time educational experience into an ongoing relationship with the cultural heritage of Bologna.